Assignment 1 – Database Design & Relationships

1. What is a database?

A database is an organized collection of data stored electronically for easy access, management, and retrieval.

2. What is a primary key?

A primary key is a column that uniquely identifies each record in a table. It cannot contain NULL values.

3. What is a foreign key?

A foreign key is a column that creates a relationship between two tables by referring to the primary key of another table.

4. What is the purpose of the Enrollment table?

The Enrollment table connects Students and Courses and manages many-to-many relationships.

5. Explain one-to-many and many-to-many relationships.

- One-to-many: One instructor teaches many courses.
- Many-to-many: Many students enroll in many courses.

6. What is SQL?

SQL (Structured Query Language) is used to create, manage, and query relational databases.

7. Difference between MySQL and PostgreSQL?

- MySQL: Faster and simple for web applications.
- PostgreSQL: More advanced features and better for complex queries.

8. Why are relationships important in databases?

Relationships reduce data redundancy and maintain data consistency.

Assignment 2 – Views and Indexing

1. What is a View?

A View is a virtual table created from one or more tables using a query.

2. Can you update data through a view?

Yes, simple views can be updated if they follow update rules.

3. What is an Index?

An Index improves data retrieval speed in a database table.

4. Benefits of Indexing?

- Faster searching
- Better query performance
- Efficient sorting and filtering

5. Difference between Table and View?

- Table stores actual data.
- View stores only query results virtually.

Assignment 3 – SQL Queries, GROUP BY, HAVING

1. Difference between WHERE and HAVING?

- WHERE filters rows before grouping.
- HAVING filters groups after GROUP BY.

2. How does ORDER BY affect performance?

Sorting large datasets may slow query execution.

3. Explain GROUP BY with example.

GROUP BY combines rows with similar values.
Example:

SELECT department, COUNT(*)
FROM Employees
GROUP BY department;

4. What is a nested subquery and why use it?

A subquery is a query inside another query used for complex filtering or calculations.

5. Can a subquery return multiple values?

Yes, using `IN` operator.

6. How can grades (A, B, C) be converted into numeric values?

Using CASE statement.

CASE grade
WHEN 'A' THEN 10
WHEN 'B' THEN 8
END

7. Difference between INNER JOIN and LEFT JOIN?

- INNER JOIN returns matching rows only.
- LEFT JOIN returns all left table rows and matching right rows.

Assignment 4 – SQL JOINS

1. What is the purpose of SQL JOINs?

JOINs combine data from multiple tables.

2. Difference between INNER JOIN and LEFT JOIN?

INNER JOIN returns only matched records, while LEFT JOIN returns all left table records.

3. When should RIGHT JOIN be used?

When all records from the right table are required.

4. Why does MySQL not support FULL OUTER JOIN directly?

MySQL uses UNION of LEFT and RIGHT JOIN instead.

5. How can FULL OUTER JOIN be simulated in MySQL?

Using:

LEFT JOIN
UNION
RIGHT JOIN

6. What is a foreign key and why is it important in joins?

It links related tables and maintains referential integrity.

7. How do joins affect query performance?

Complex joins on large tables may slow queries without indexing.

Assignment 5 – Procedures and Functions

1. What is PL/SQL?

PL/SQL is Oracle's procedural extension of SQL.

2. Difference between procedure and function?

- Procedure performs operations.
- Function returns a value.

3. Can a function be used inside a SELECT statement?

Yes.

4. Why is COMMIT used in procedures?

To permanently save transaction changes.

5. What happens if a function does not return a value?

It generates an error.

6. Can procedures accept parameters?

Yes, IN, OUT, and IN OUT parameters.

7. Where are PL/SQL procedures used in real-world applications?

Banking systems, payroll systems, ERP applications, etc.

Assignment 6 – Triggers and Cursors

1. What is a trigger in PL/SQL?

A trigger automatically executes when INSERT, UPDATE, or DELETE occurs.

2. Difference between BEFORE and AFTER trigger?

- BEFORE trigger executes before the event.
- AFTER trigger executes after the event.

3. What is a cursor?

A cursor processes query results row by row.

4. Why are cursors required in PL/SQL?

For row-wise processing and complex logic.

5. What is the use of :OLD and :NEW keywords?

- :OLD stores previous values.
- :NEW stores updated values.

6. Can triggers be rolled back?

Yes, if transaction rollback occurs.

7. Difference between implicit and explicit cursors?

- Implicit: Automatically created by Oracle.
- Explicit: Created manually by programmer.

Assignment 7 – MongoDB CRUD & Aggregation

1. What is NoSQL database?

A non-relational database used for unstructured or semi-structured data.

2. What is MongoDB?

MongoDB is a document-oriented NoSQL database.

3. Difference between document and relational databases?

- Document DB stores JSON-like documents.
- Relational DB stores tables and rows.

4. What is aggregation pipeline?

A framework for data processing and analysis in MongoDB.

5. What is indexing and why is it required?

Indexing speeds up search operations.

6. What is BSON?

Binary JSON format used by MongoDB.

7. How does indexing improve query performance?

It reduces collection scanning and speeds data retrieval.

Assignment 8 – Social Media Data Processing Using MongoDB

1. Why is NoSQL suitable for social media data?

Because social media data is large, flexible, and unstructured.

2. What is $unwind in MongoDB aggregation?

`$unwind` separates array elements into individual documents.

3. How are trending hashtags identified?

Using aggregation queries like GROUP and COUNT.

4. What is the advantage of indexing in MongoDB?

Faster query execution.

5. Difference between SQL and NoSQL databases?

- SQL uses tables and fixed schema.
- NoSQL uses flexible documents.

6. What is BSON?

Binary format of JSON used in MongoDB.

Assignment 9 – MySQL Backup and Recovery

1. What is database backup?

A copy of database data stored for recovery purposes.

2. Why is backup important in DBMS?

To prevent data loss due to failures or cyberattacks.

3. What is mysqldump?

A MySQL utility used for logical database backup.

4. Difference between logical and physical backup?

- Logical backup stores SQL statements.
- Physical backup stores actual database files.

5. What happens if backup is not taken regularly?

Data loss and recovery problems may occur.

6. How do you restore a MySQL database?

Using MySQL restore command:

mysql -u root -p database_name < backup.sql

7. What is disaster recovery?

The process of restoring systems and data after failure or disaster.